# Module 7: Developing Event-Driven Serverless Solutions

AWS Academy Cloud Developing

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# Section 1: Introduction

## Module 7: Developing Event-Driven Serverless Solutions

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# Module objectives

At the end of this module, you should be able to:

- Explain serverless computing
- Describe how AWS Lambda works
- Recognize Lambda invocation models
- Identify how to use AWS Identity and Access Management (IAM) to grant Lambda permissions
- Indicate the steps to author and configure a Lambda function
- Explain how to deploy Lambda functions
- Identify why AWS X-Ray is a critical developer tool
- Create Lambda functions

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# Module overview

## Sections

1. Introduction
2. Introducing serverless computing
3. Introducing Lambda
4. Invoking Lambda functions
5. Setting permissions for Lambda
6. Authoring and configuring Lambda functions
7. Deploying Lambda functions
8. Monitoring and debugging tools for application developers

## Demonstration

Using X-Ray with Lambda

## Lab

Creating Lambda Functions Using the AWS SDK for Python

Knowledge check

aws

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# Café business requirement

Sofía finished using the mock endpoints to test the API, and now she wants to launch the dynamic version of the café website. She needs to update the API integration to connect to the backend database.

```
cursor.close()

sql = "INSERT INTO hive (dev_id, humidity, temperature) VALUES (%s, %s, %s)"

def on_connect(client, userdata, flags, rc):
    print("Connected with result code " + str(rc))

    client.subscribe("+/devices/+/up")

def on_message(client, userdata, msg):
    print(msg.topic + " 666 " + str(msg.payload)+"\n")
    data = json.loads(msg.payload.decode('utf8').replace("'", '"'))
    val = (data["dev_id"], data["payload_fields"]["humidity"], data["payload_fields"]["temperature"])
    connection = mysql.connector.connect(
        host="bees-mariaDB",
        user="bees",
        passwd=mysql_password,
        database="bees"
    )
    cursor = connection.cursor()
    cursor.execute(sql, val)
    connection.commit()
    print(mycursor.rowcount, "record inserted.")
    cursor.close()
    connection.close()

client = mqtt.Client()
```

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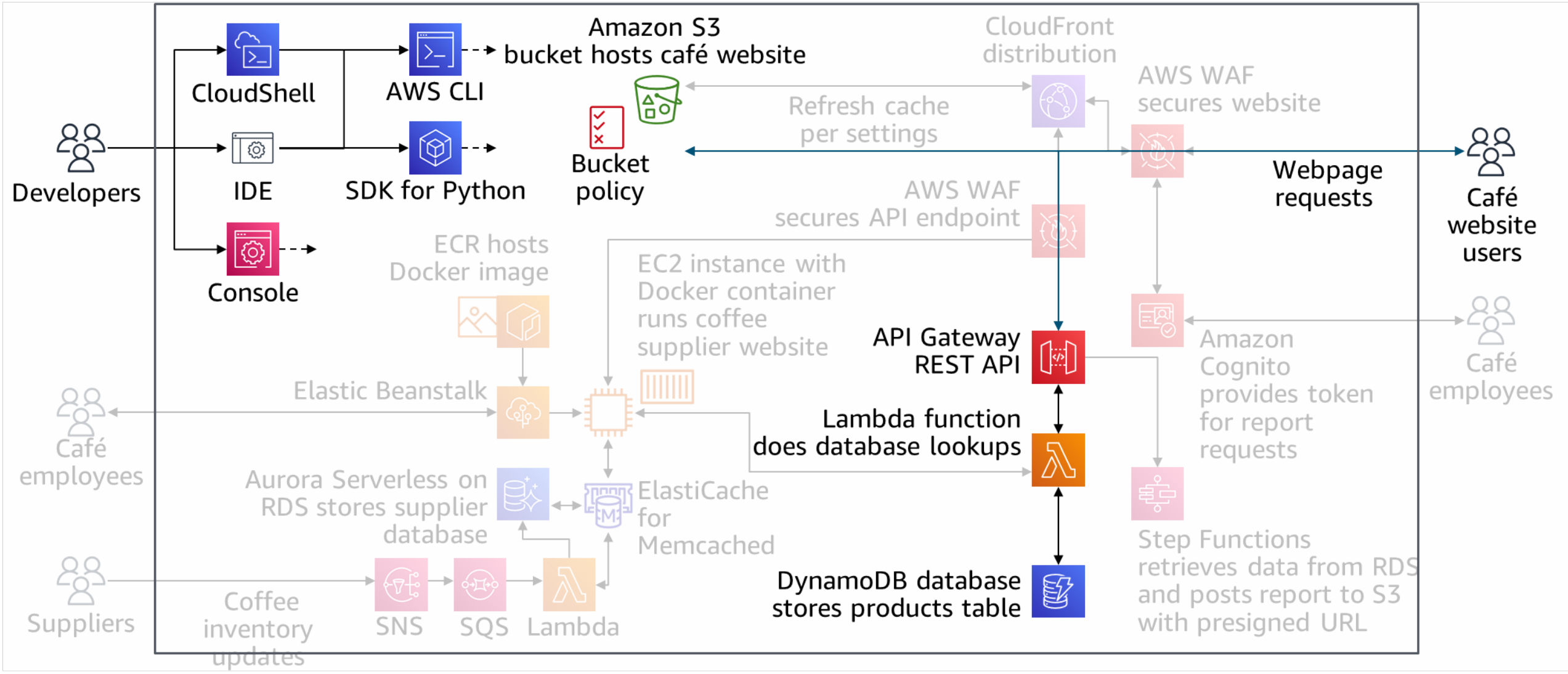
# Serverless as part of developing a cloud application

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# Section 2: Introducing serverless computing

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# Serverless computing

Level of abstraction

Focus on business logic

## On premises

- Heavy capital expense
- Guessed capacity
- Deployed in months
- Years of maintenance
- Low innovation factor

## Virtual servers in the cloud

- Hardware independence
- Elastic resources and scale
- Faster provisioning
- Reduced maintenance

## Containers

- Platform independence
- Higher resource utilization
- Easier and faster deployments
- Isolation and sandboxing

## Serverless

- No OS instances to manage
- Flexible scaling
- Pay per usage
- Built-in fault tolerance
- Zero maintenance

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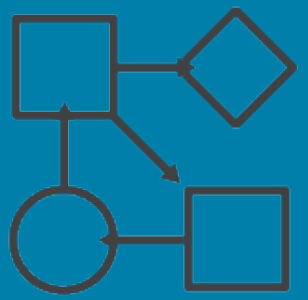
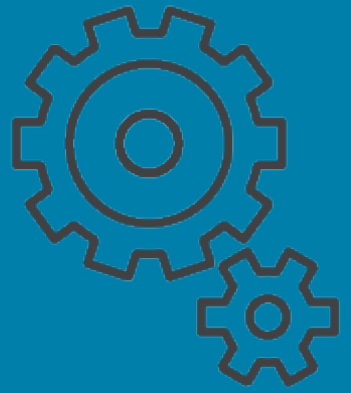
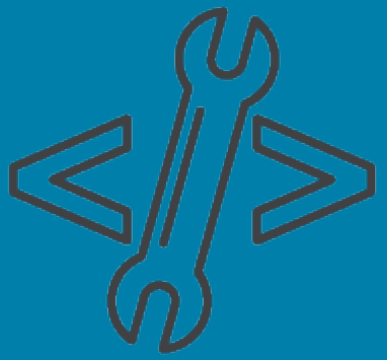
# Building modern serverless applications

| Small pieces, loosely joined | Purpose-built data stores | Specialized services for integration | Automation by using code |
| --- | --- | --- | --- |
|  | |  |  |
| Serverless compute and event-driven patterns | Serverless, purpose-built data stores | Managed services for integration | Deployment frameworks |

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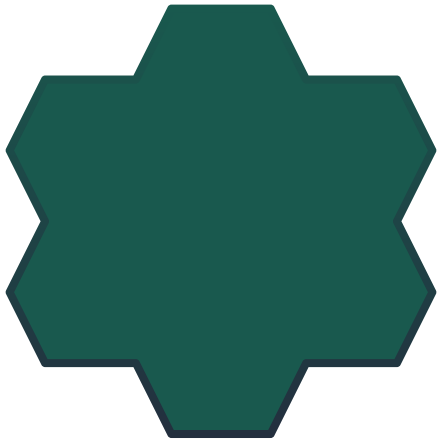
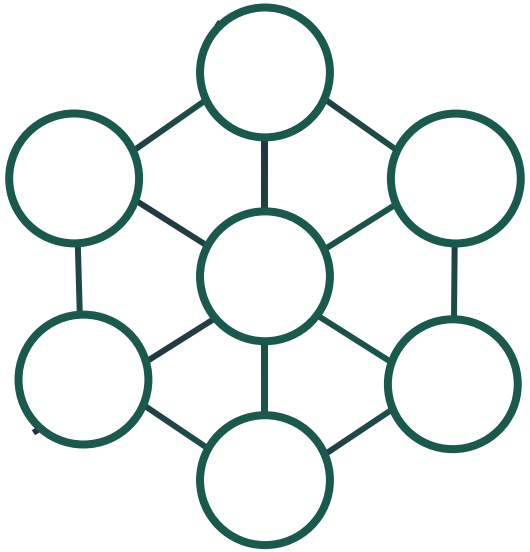
# Moving away from a monolithic application



Monolith

Does everything



Microservice

Does one thing

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# Section 3: Introducing Lambda

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# AWS Lambda

AWS Lambda

Lambda is a serverless compute service that you can use to run code without provisioning or managing servers.

- Invokes your code in response to events
- Scales automatically
- Provides built-in code monitoring and logging with Amazon CloudWatch

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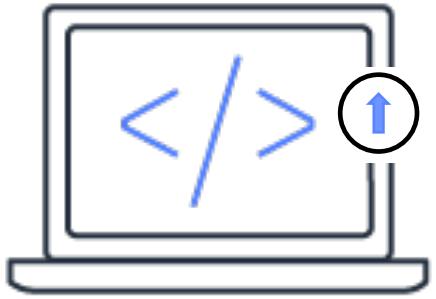
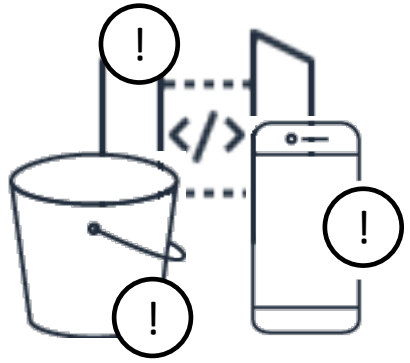
# How Lambda works



Upload your code to Lambda or write code in the Lambda editor



Set up your code to run when events occur in other AWS services, at HTTP endpoints, or as part of in-app activity

Lambda runs your code only when it is activated by an event and uses only the compute resources that are needed

Pay only for the compute time that you use

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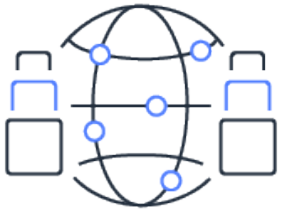
# Lambda features

Offers the ability to bring your own code

Integrates with and extends other AWS services

Offers flexible resource and concurrency models

Offers a flexible permissions model



Provides built-in availability and fault tolerance

Reduces the need to pay for idle resources

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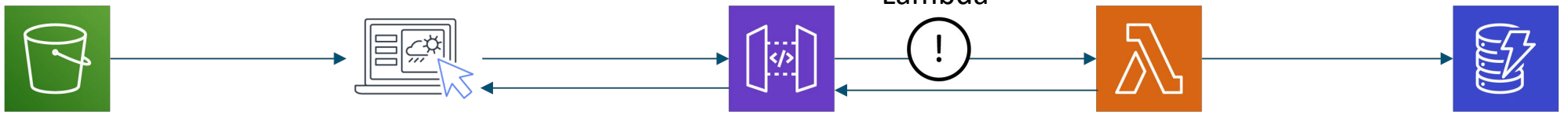
# Example use cases for Lambda

Web app




Amazon S3

The front-end code for the weather app is hosted on Amazon S3

Website

A user chooses a link for local weather information

API Gateway

The app makes a REST API call to the endpoint

Lambda

Lambda runs code to get the weather data from DynamoDB and returns the data

DynamoDB

DynamoDB contains the weather data that the app uses

Real-time stream processing

Kinesis

A social media stream is loaded into Kinesis in real time

Lambda

Lambda runs code that generates hashtag trend data

DynamoDB

Hashtag trend data is stored in DynamoDB

Social media trend data is immediately available for business users to query

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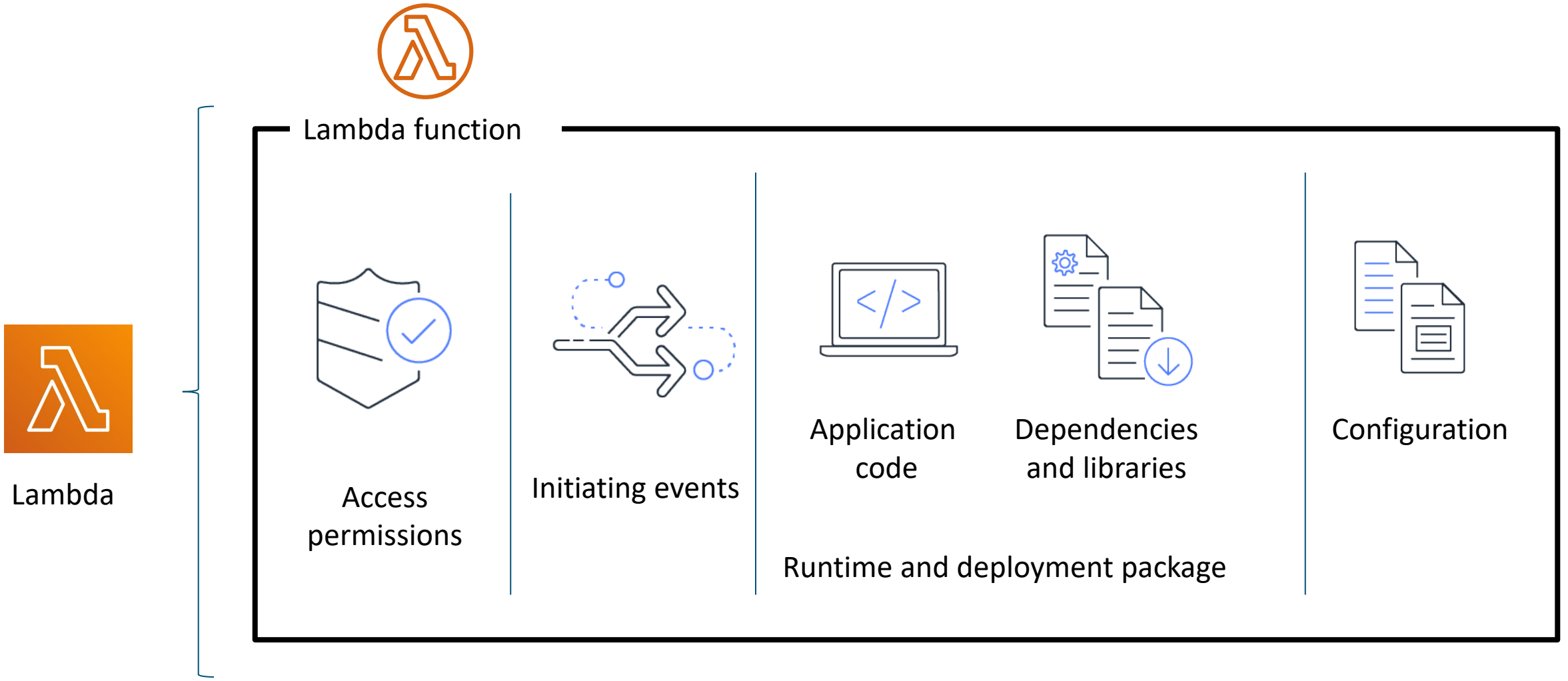
# Lambda functions

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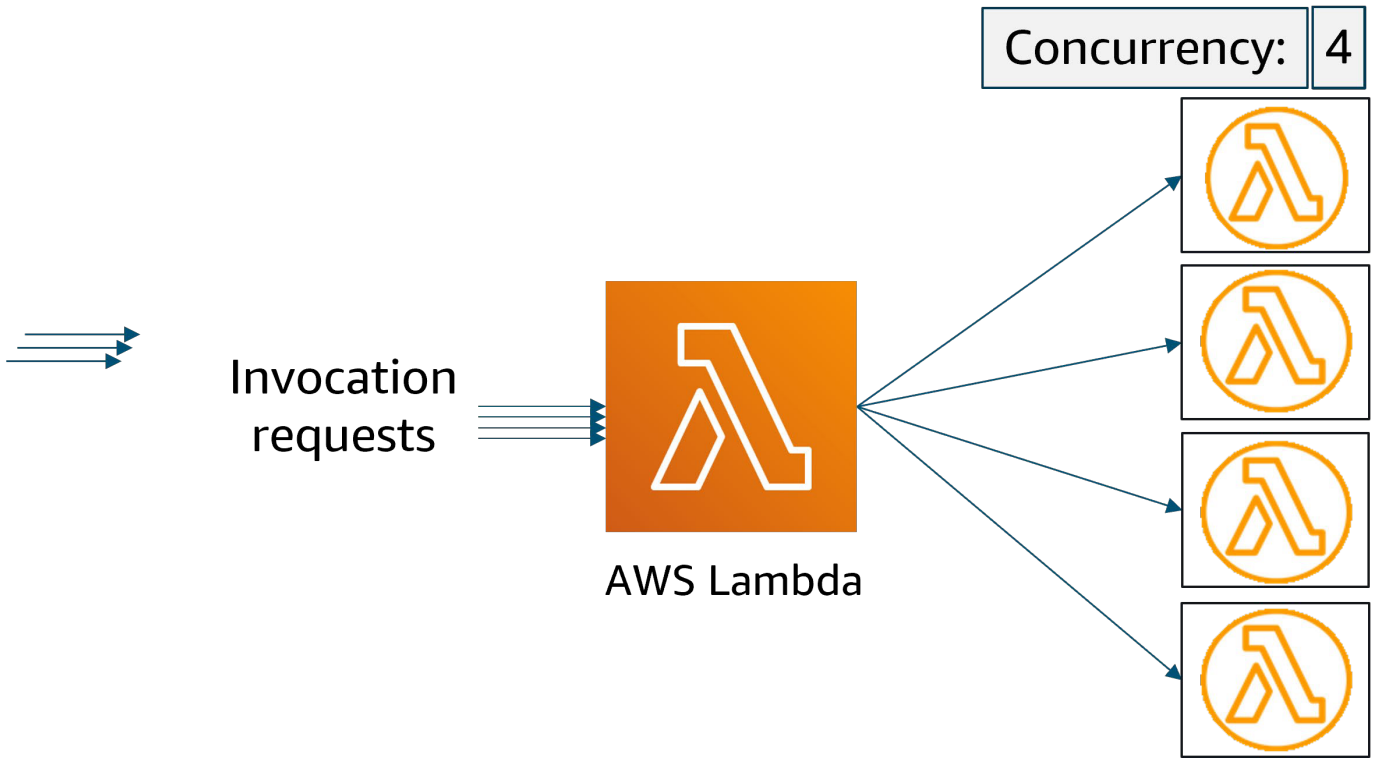
# Lambda functions run in on-demand ephemeral environments

**Concurrency:** The number of function invocations that are running at one time

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# Section 3 key takeaways

- Lambda is a serverless compute service that integrates with other AWS services.
- You can use Lambda for many use cases including web applications, backends, and data processing.
- Lambda functions run in on-demand temporary environments.

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# Section 4: Invoking Lambda functions

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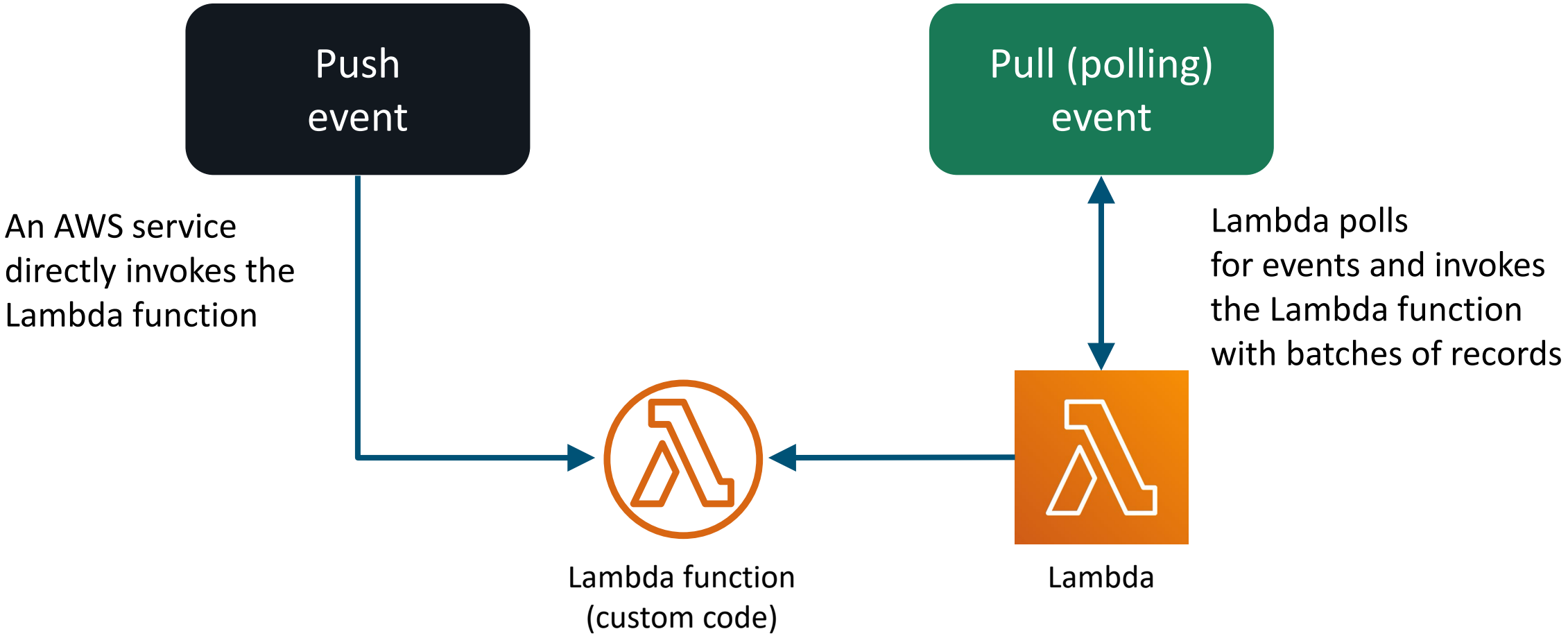
# Push and pull models

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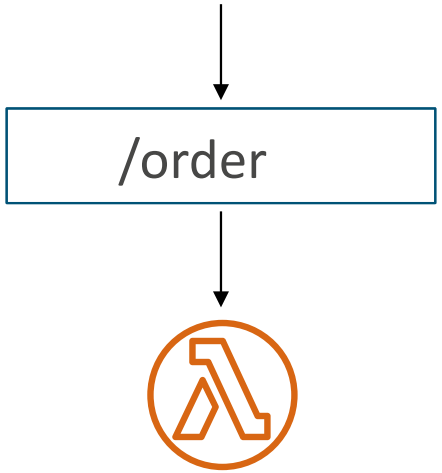
# Push event types

## Synchronous

The other service waits for a response from your function.

API Gateway




No retry built in

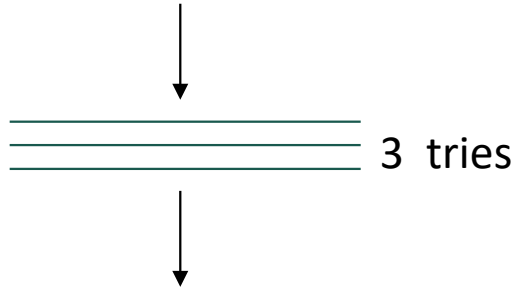
Invoke via API:
*RequestResponse*

Lambda function

## Asynchronous

Lambda queues the event before passing it to your function.

Amazon S3



0–2 retries

3 tries

Invoke via API:
*Event*

Lambda function

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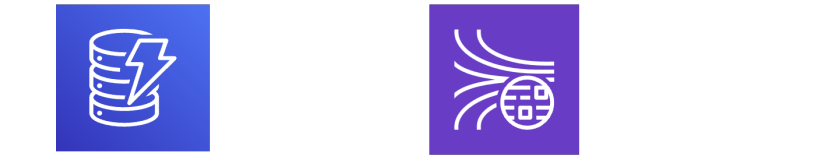
# Pull (polling) event types

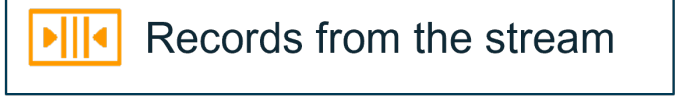
## Lambda polls a stream



Amazon DynamoDB Streams



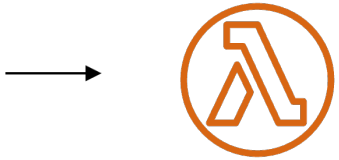
Amazon Kinesis Data Streams



shard

Errors in a shard block further processing

Lambda



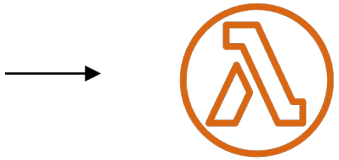
Lambda function

## Lambda polls a queue

Amazon Simple Queue Service (Amazon SQS)

Errors in a batch are returned to the queue

Lambda



Lambda function

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# Invoking a function synchronously

Invoke

```
aws lambda invoke \
--function-name my-function \
--payload '{ "name": "Bob" }' \
response.json
```

Response

```
{"ExecutedVersion": "$LATEST","StatusCode": 200}
```

aws

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# Invoking a function asynchronously

Invoke

```
aws lambda invoke \
--function-name my-function \
--invocation-type Event \
--payload '{ "name": "Bob" }' \
response.json
```

Ack

```
{"StatusCode": 202}
```

aws

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# Section 4 key takeaways

- Lambda invokes Lambda functions by using one of these invocation models:
  - Synchronous
  - Asynchronous
  - Polling (event source mapping)
- Lambda has built-in handling that differs by event source type.

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# Section 5: Setting permissions for Lambda

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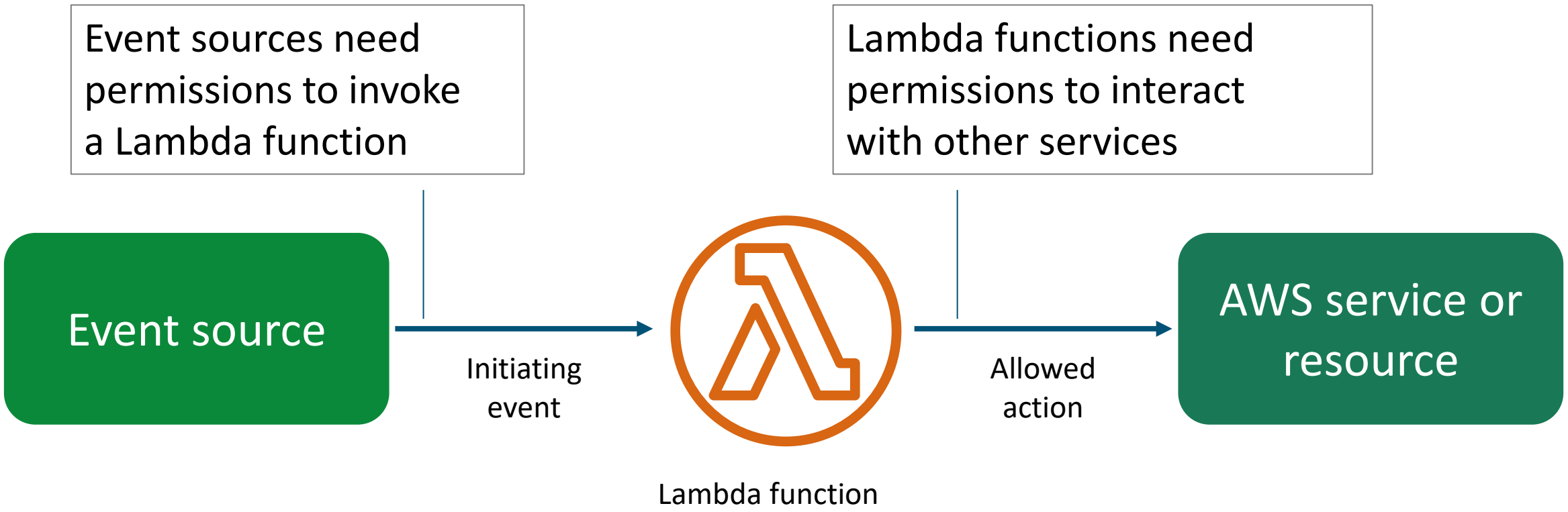
# Lambda permissions

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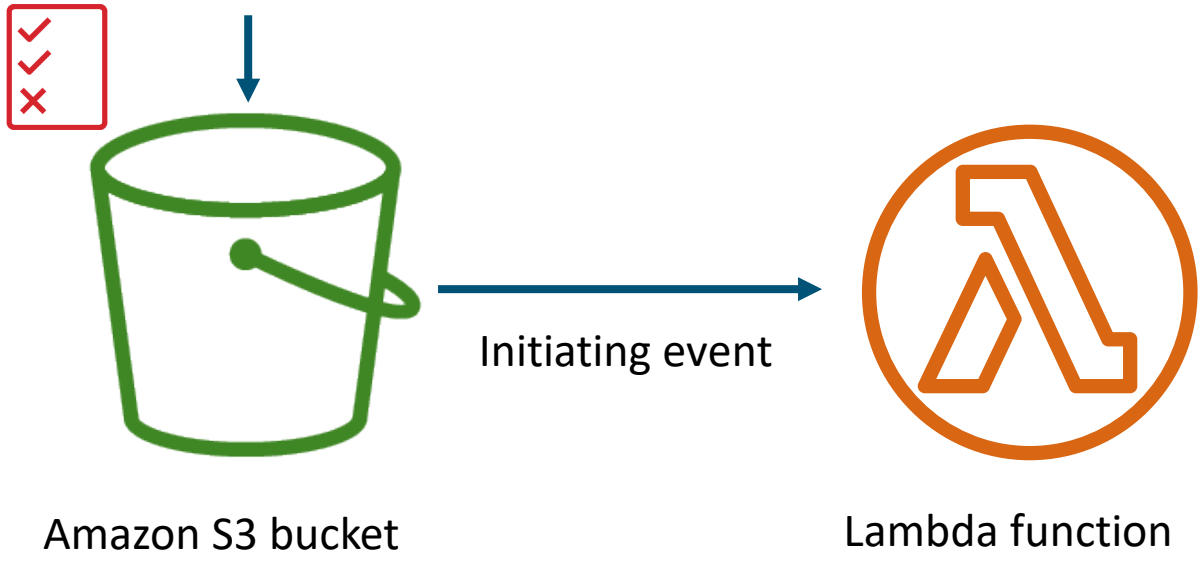
# Invocation permissions

An IAM resource policy gives permissions to invoke the function.




Resource policy:

- Associated with synchronous or asynchronous event source
- Allows the event source to take the *lambda:InvokeFunction* action

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# Example resource (function) policy

```
{
        "Version": "2012-10-17",
        "Id": "default",
        "Statement": [
                {
                        "Sid": "lambda-fd269e28-988b-4d2b-96ae-eabcd7dc399c",
                        "Effect": "Allow",
                        "Principal": {
                                "Service": "s3.amazonaws.com"
                        },
                        "Action": "lambda:InvokeFunction",
                        "Resource": "arn:aws:lambda:function:myFirstFunction",
                        "Condition": {
                                "ArnLike": {
                                        "AWS:SourceARN": "arn:aws:s3:::myBucket1"
                                }
                        }
                }
        ]
}
```

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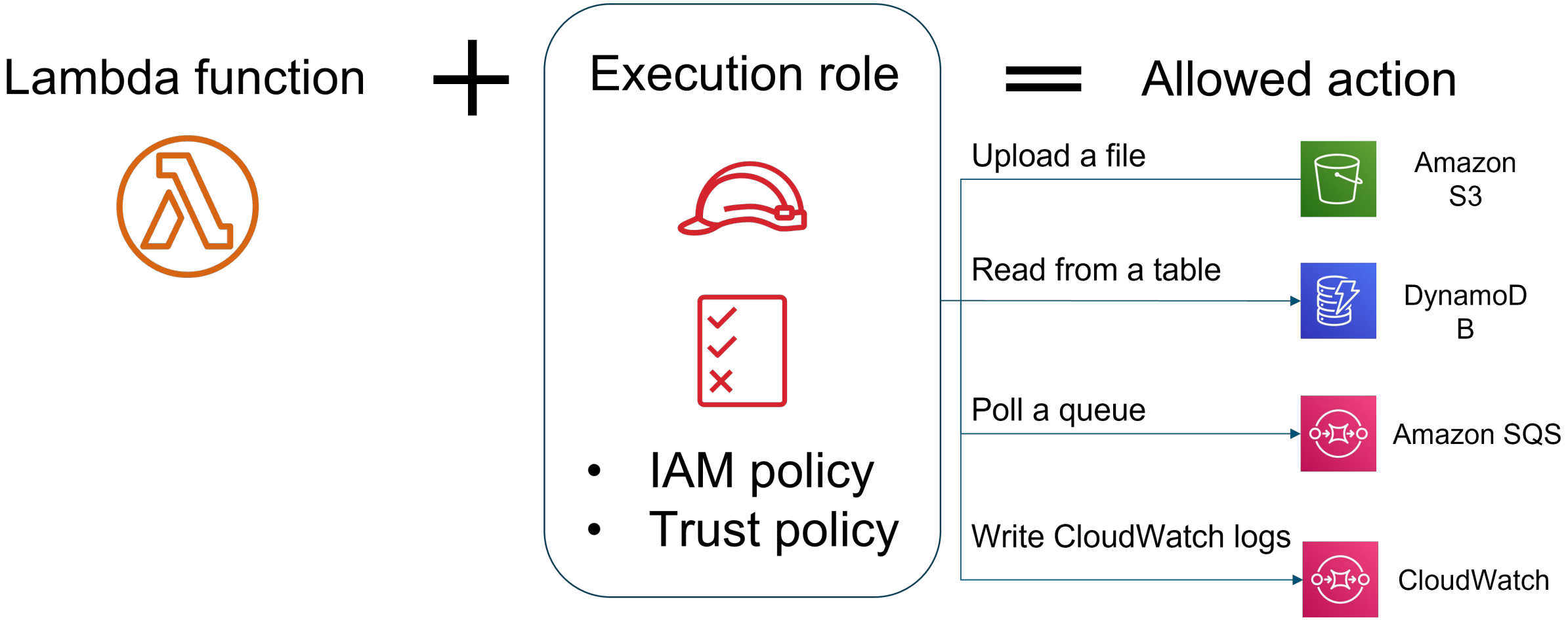
# Lambda execution

The Lambda execution role specifies what the Lambda function is permitted to do.

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# Execution role example

IAM policy

```
{
        "Effect": "Allow",
        "Action": [
                "logs:CreateLogGroup",
                "logs:CreateLogStream",
                "logs:PutLogEvents"
        ],
        "Resource": "arn:aws:logs:*:*:*"
}
```

Trust policy

```
{
    "Effect": "Allow",
    "Principal": {
                "Service": "lambda.amazonaws.com"
      },
    "Action": "sts:AssumeRole"
}
```

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# Adding permissions by using the AWS CLI

Give Amazon SNS permissions to invoke a function

```
aws lambda add-permission \
--function-name my-function \
--action lambda:InvokeFunction \
--statement-id sns \
--principal sns.amazonaws.com
```

Output

```
{
"Statement":
{
"Sid":"sns",
"Effect":"Allow",
"Principal":{
"Service":"sns.amazonaws.com"
},
"Action":"lambda:InvokeFunction",
"Resource":"arn:aws:lambda:us-east-2:123456789012:function:my-function"
}
}
```

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# Section 5 key takeaways

- Lambda functions require permissions to be invoked and permissions to interact with other services.
- Both types of permissions are defined by using IAM or trust policies.

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# Section 6: Authoring and configuring Lambda functions

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# The function handler

Handler (event, context) method

## Event object

- Passes event information to the handler
- Uses a predefined object format for AWS integrations and events
- Can be tested with user-defined custom objects

## Context object

- Passes runtime information to the handler
- Includes, at a minimum, these methods or properties:
  - awsRequestId
  - getRemainingTimeInMillis()
  - logStreamName

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# Lambda function handler example

event object

```
{
    "first_name": "John",
    "last_name": "Smith"
}
```

Handler

```
def lambda_handler(event, context):
    message = 'Hello {} {}!'.format(event['first_name'],
                                    event['last_name'])
    return {
        'message' : message
    }
```

response

```
{
    "message": "Hello John Smith!"
}
```

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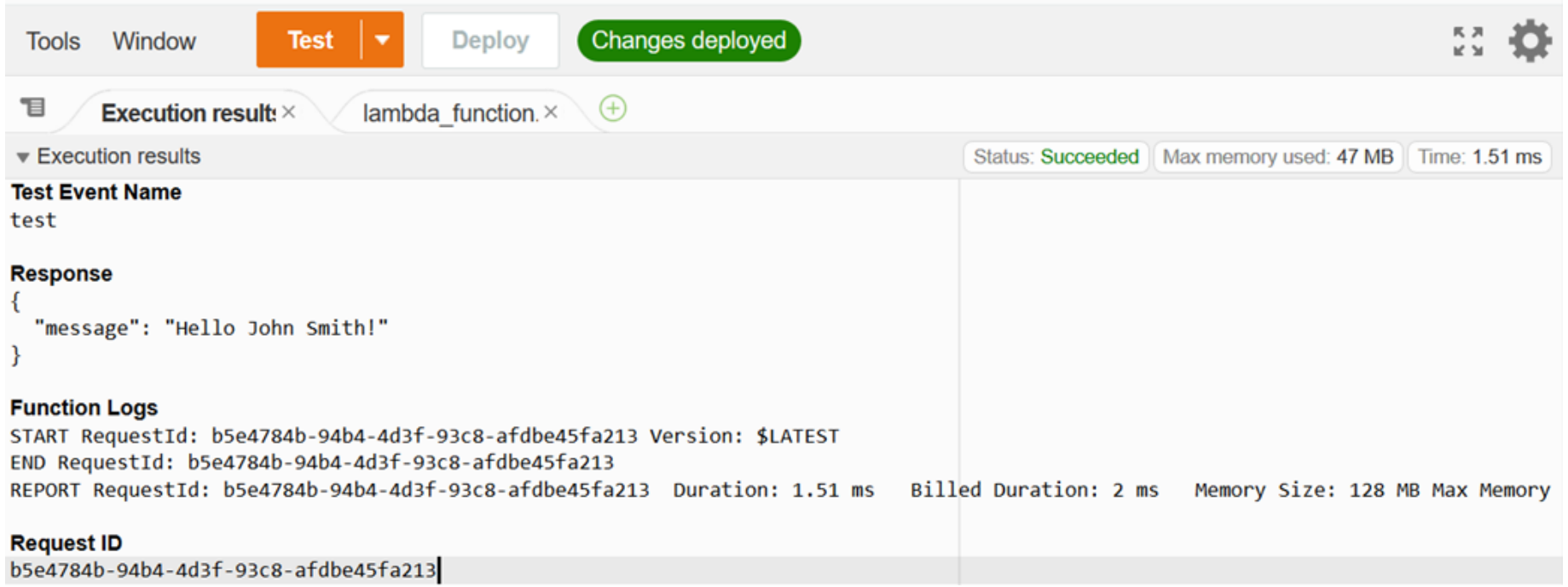
# Example testing from the console

Create test event

Review results

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# Performance-related configurations

| Configuration | Description |
|---|---|
| Memory | Set the amount of memory and proportional CPU that is allocated to the function. Lambda allocates CPU power linearly in proportion to the memory that you set. Values from 128 MB–10,240 MB. |
| Timeout | Choose the maximum amount of time to let a function run before ending it if it has not completed. Values from 1 second up to 15 minutes. |
| Concurrency | Number of invocations of a function that can run at the same time. By default, you can have 1,000 concurrency invocations per Region for an account. This limit is a soft limit. Per function, you can set a limit to prevent overwhelming a downstream system or to reserve capacity from within the account pool for the function. |
| Provisioned concurrency | Number of Lambda environments to *keep warm*. When Lambda creates the temporary environment for your function, it will attempt to reuse it for additional invocations. This approach saves startup time. When Lambda must initialize new environments to run a function, an invocation might get a *cold start*, which creates startup latency. Use provisioned concurrency to avoid cold starts. Provisioned concurrency is priced separately. |
| Monitoring and operations | Settings to enable X-Ray (active tracing) and CloudWatch Lambda Insights (collect and aggregate Lambda function runtime performance metrics and logs). |

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# Resource-related configurations

| Configuration | Description |
|---|---|
| Triggers | Event sources that invoke a function. |
| Permissions | The resources that have permissions to invoke the function and the permissions that the function has to interact with other resources. |
| Destinations | An SNS topic, SQS queue, other Lambda function, or EventBridge event bus. It receives invocation records from a function when it is successful (on success) or when it fails (on failure). |
| Asynchronous invocation | Settings for number of retry attempts on async invocations (0–2) and how long to keep an event waiting to be invoked (up to 6 hours). Also, configuration of a dead-letter queue for functions that continue to fail. |
| VPC | Settings to enable your Lambda function to access resources in a custom VPC. A custom VPC defines a private network of resources, such as databases, cache instances, or internal services. |
| State machines | Step Functions state machines that can invoke your function directly in at least one of the steps. |
| Database proxies | Settings that are used when connecting to an Amazon Relational Database Service (Amazon RDS) instance. These settings are used to set up a database proxy to manage connection pooling for database connections. |
| File systems | Settings to connect an Amazon Elastic File System (Amazon EFS) file system so that the function can access the file system at runtime. |

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# Code-related configurations

| Configuration | Description |
| --- | --- |
| Runtime | Runtime that the function will use or language that the code will be written in. Choose from supported runtimes that are listed in the AWS Lambda Developer Guide or use a custom runtime. |
| Environment variables | Key-value pairs that are accessible from your function code. Environment variables are useful to store configuration settings without the need to change function code. |
| Tags | Labels that you assign to an AWS resource. Each tag consists of a key and an optional value. You can use tags to search and filter your resources or track your AWS costs. |
| Code signing | Option to ensure that code has been signed by an approved source and has not been altered. |

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# Design a Lambda function: Best practices for a Lambda environment

- Treat functions as stateless.
- Include only what you need.
- Reuse the temporary runtime environment.

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# Design a Lambda function: Best practices for writing code

- Separate the core business logic (outside of the handler method).
- Write modular functions.
- Include logging statements.
- Include results information.
- Use environment variables.
- Avoid recursive code.
- Don't call one function from another.

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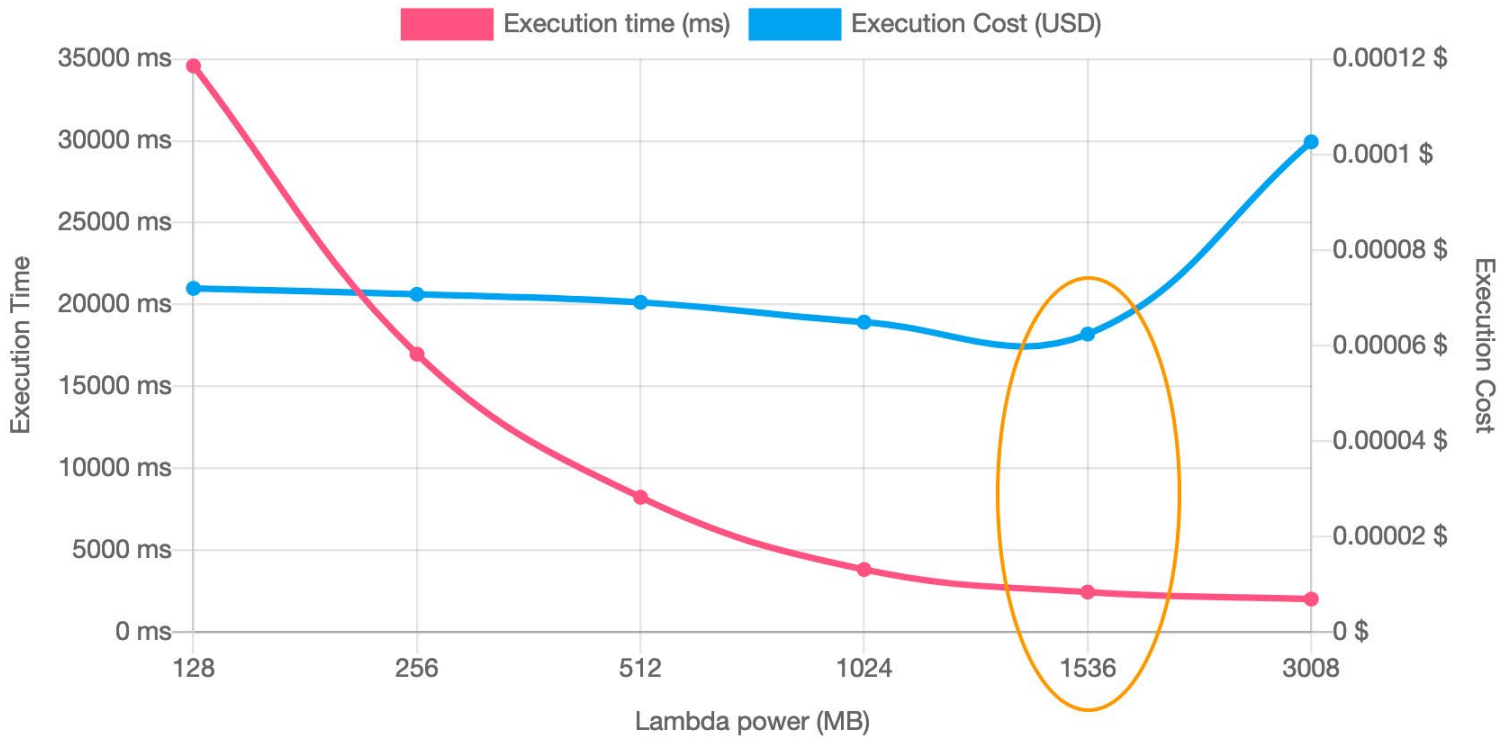
# Tuning your function

- Higher memory configurations have a higher per-1-millisecond cost but might decrease duration costs and concurrency needs.
- Find the balance that optimizes for speed and cost.

Example results for two CPU-intensive functions, which become cheaper ***and*** faster with more power




Source: https://github.com/alexcasalboni/aws-lambda-power-tuning

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# Section 6 key takeaways

- Lambda supports different languages and runtimes.
- The function handler is the entry point into the Lambda function.
- You can configure attributes including the amount of memory and function timeout.
- Follow best coding practices to create efficient Lambda functions.

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# Section 7: Deploying Lambda functions

## Module 7: Developing Event-Driven Serverless Solutions

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# Deployment options

## .zip archive

- Choose a **runtime** when you **create the function.**
- Compress files for application code and dependencies, and **upload them to Lambda.**

## Container image

- Choose a **runtime and linux distribution** when **creating the image.**
- Package your code and dependencies as a **container image**.
- **Upload** the image to your **container registry** that is hosted on Amazon Elastic Container Registry (Amazon ECR).

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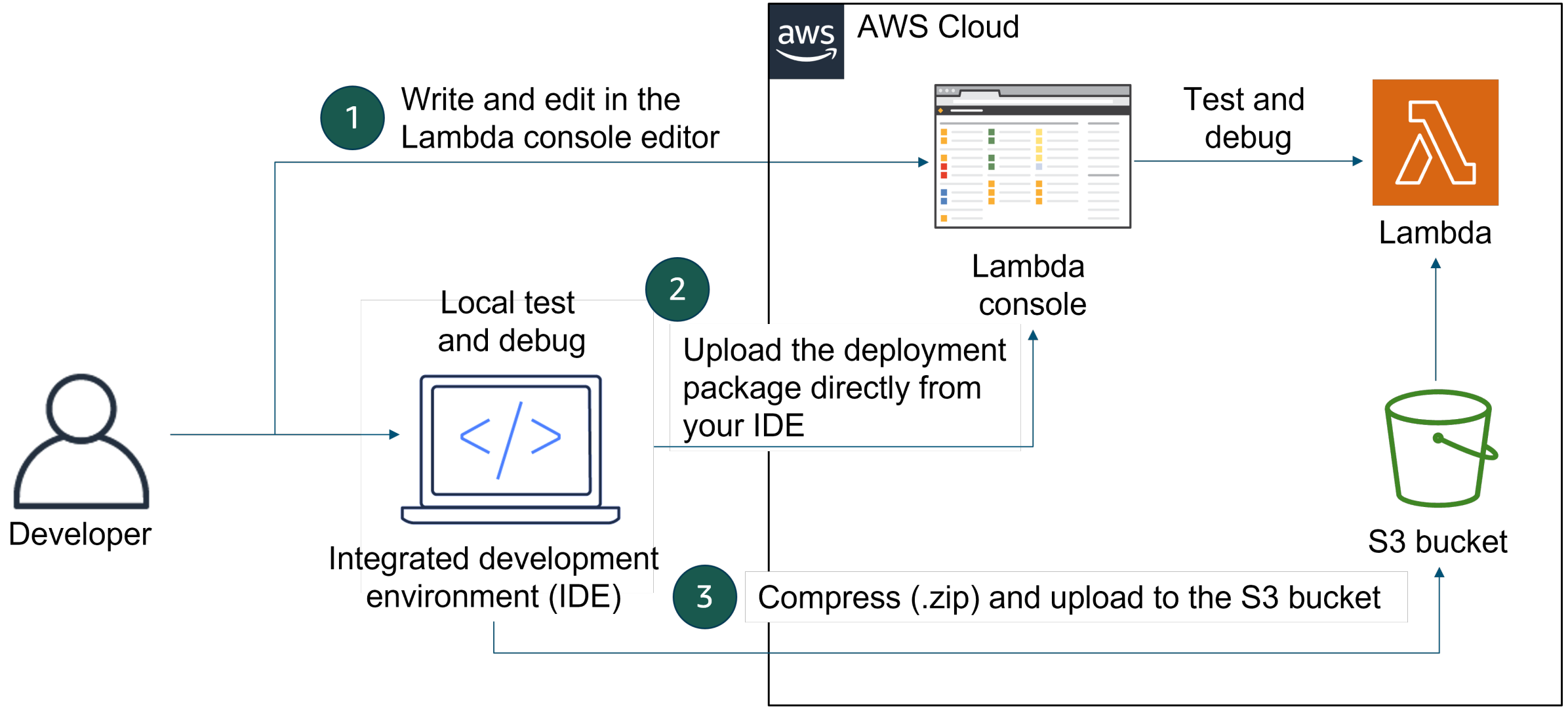
# Uploading .zip deployment packages

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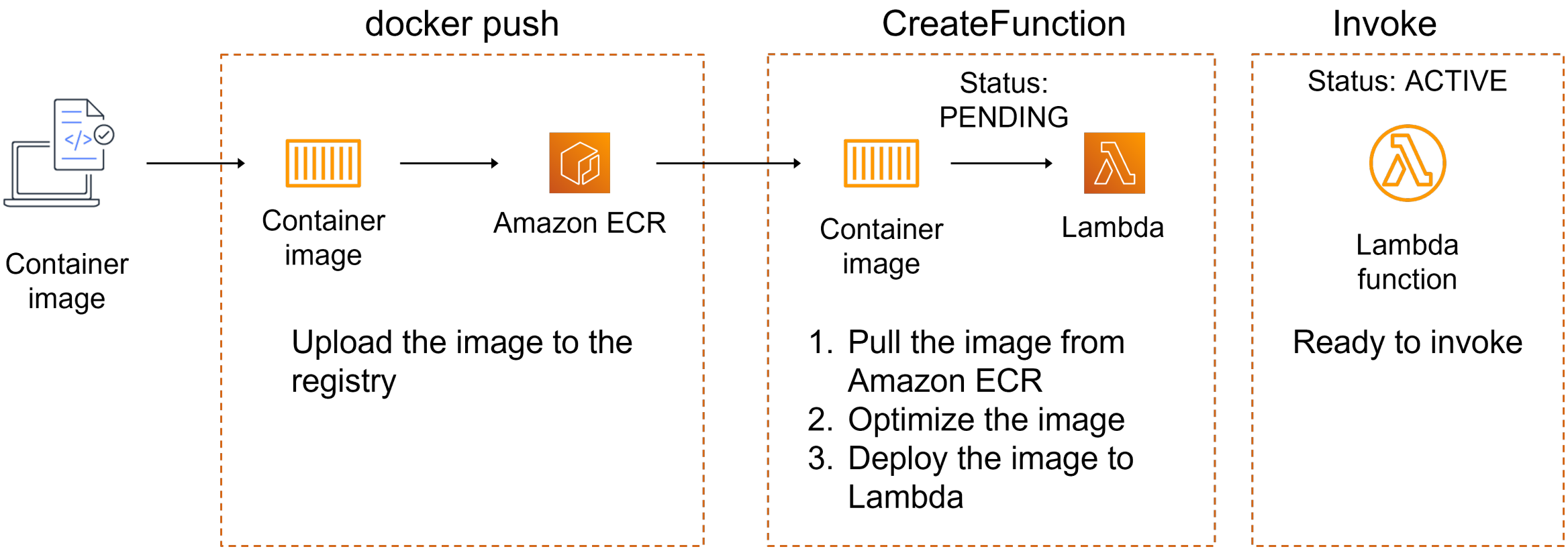
# Deploying as a container image

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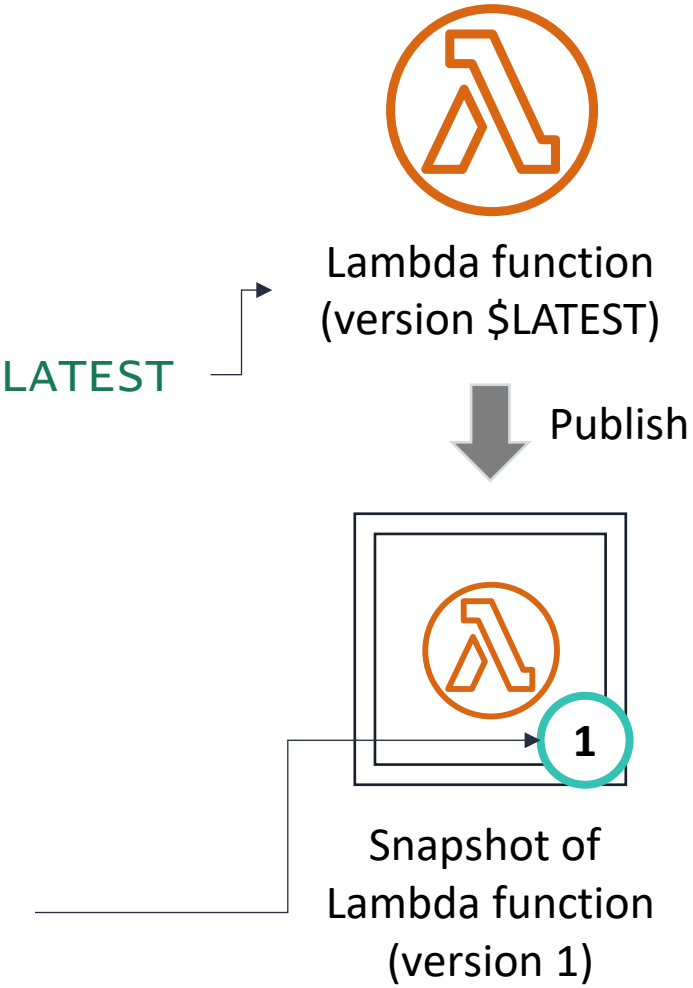
# Versioning

- Immutable copies of Lambda function code and configuration
- Create: $LATEST version

```
arn:aws:lambda:aws-region:acct-id:function:helloworld:$LATEST
```

- Publish: Snapshot copy of $LATEST
- Each version has its own Amazon Resource Name (ARN) with a new sequential version number

```
arn:aws:lambda:aws-region:acct-id:function:helloworld:1
```

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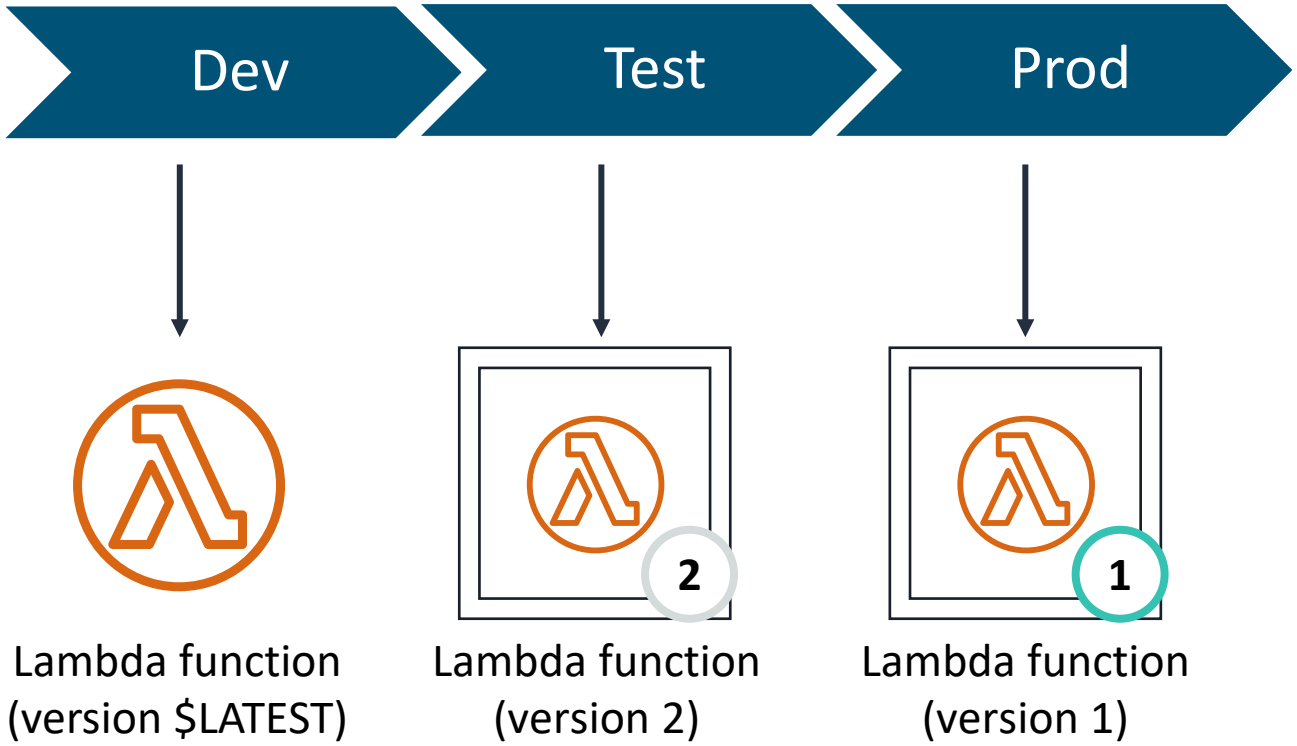
# Aliases

- Mutable pointers to versions
- Each alias has its own ARN




Lambda function test alias:
`arn:aws:lambda:aws-region:acct-id:function:helloworld:test`

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# Example of using versioning and aliases

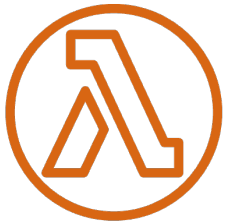
## Without aliases:

Amazon S3

arn:aws:lambda:aws-region:acct-id:function:helloworld:$LATEST

arn:aws:lambda:aws-region:acct-id:function:helloworld:*1*

Event source mapping that uses the Lambda function ARN (update with each version)



Lambda function (version $LATEST)

Publish

## With aliases:

Amazon S3

Event source mapping that uses the alias ARN (don't need to update with each version)

arn:aws:lambda:aws-region:acct-id:function:helloworld:PROD

Snapshot of Lambda function (version 1)

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# Custom runtimes

- With custom runtimes you can use **other languages** in Lambda.
- Lambda provides a **runtime interface** for getting events and sending responses.
- The **bootstrap file** is the runtime's entry point.
- You can deploy a custom runtime next to your function code or **in a layer**.

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# Lambda layers

Centrally manage code and data that are shared across multiple functions

- Reduce the size of deployment packages
- Speed up deployments
- Limits:
  - Up to five layers
  - 250 MB

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# Example deploying from the AWS CLI

Create function

```
aws lambda create-function \
--function-name my-function \
--runtime nodejs10.x \
--zip-file fileb://my-function.zip \
--handler my-function.handler \
--role arn:aws:iam::123456789012:role/service-role/MyTestFunction-role-tges6bf4
```

Output

```
{
"TracingConfig": {
"Mode": "PassThrough"
},
"CodeSha256": "PFn4S+er27qk+UuZSTKEQfNKG/XNn7QJs90mJgq6oH8=",
"FunctionName": "my-function",
"CodeSize": 308,
"RevisionId": "873282ed-4cd3-4dc8-a069-d0c647e470c6",
"MemorySize": 128,
"FunctionArn": "arn:aws:lambda:us-west-2:123456789012:function:my-function",
"Version": "$LATEST",
"Role": "arn:aws:iam::123456789012:role/service-role/MyTestFunction-role-zgur6bf4",
"Timeout": 3,
"LastModified": "2019-08-14T22:26:11.234+0000",
"Handler": "my-function.handler",
"Runtime": "nodejs10.x",
}
```

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# Example of updating a function from the AWS CLI

Update function

```
aws lambda update-function-code \
--function-name my-function \
--zip-file fileb://my-function.zip
```

Output

```
{
"FunctionName": "my-function",
"LastModified": "2019-09-26T20:28:40.438+0000",
"RevisionId": "e52502d4-9320-4688-9cd6-152a6ab7490d",
"MemorySize": 256,
"Version": "$LATEST",
"Role": "arn:aws:iam::123456789012:role/service-role/my-function-role-uy3l9qyq",
"Timeout": 3,
"Runtime": "nodejs10.x",
"TracingConfig": {
"Mode": "PassThrough"
},
"CodeSha256": "5tT2qgzYUHaqwR716pZ2dpkn/0J1FrzJmlKidWoaCgk=",
"CodeSize": 304,
"FunctionArn": "arn:aws:lambda:us-west-2:123456789012:function:my-function",
"Handler": "index.handler"
}
```

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# Section 7 key takeaways

- Lambda has built-in versioning so that you can create immutable versions of your code.
- You can use aliases to point at different versions of your Lambda function.
- Lambda provides popular runtimes, but you can create custom runtimes.
- Lambda layers are libraries of code that you create and which other Lambda functions can call.

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# Section 8: Monitoring and debugging tools for application developers

## Module 7: Developing Event-Driven Serverless Solutions

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# Modern applications challenge traditional monitoring approaches

Short-lived resources

More devices, services, and data

Faster release velocity

User experience

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# Amazon CloudWatch

Amazon CloudWatch

CloudWatch collects monitoring and operational data in the form of logs, metrics, and events.

- Get a **unified view** of AWS resources, applications, and services
- Review logs as a **flow of events ordered by time**
- **Query** log data interactively with **CloudWatch Logs Insights**
- Use **default operational metrics** from AWS services
- **Alarm** based on metric thresholds and initiate **automated actions**

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# Example of Lambda logging

Log group

Log stream

CloudWatch > Log Groups > /aws/lambda/FAQ > 2020/08/27/[$LATEST]05419d29f9014fc3b1634b1e9968ef28

| Time | Message |
|---|---|
| 19:35:32 | START RequestId: 41c8dcf0-ff10-4d96-af2a-1ab5e346c937 Version: $LATEST |
| 19:35:32 | 2020-08-27T19:35:32.553Z 41c8dcf0-ff10-4d96-af2a-1ab5e346c937 INFO … |
| 19:35:32 | 2020-08-27T19:35:32.575Z 41c8dcf0-ff10-4d96-af2a-1ab5e346c937 INFO … |
| 19:35:32 | END RequestId: 41c8dcf0-ff10-4d96-af2a-1ab5e346c937 |
| 19:35:32 | REPORT RequestId: 41c8dcf0-ff10-4d96-af2a-1ab5e346c937 Duration: 64.40 ms Billed Duration: 100 ms Memory Size: 128 MB Max Memory Used: 68 MB Init Duration: 162.85 ms |
| 19:35:36 | START RequestId: a23cbea6-3053-4b51-8721-ff11aba24e33 Version: $LATEST |
| … | |

Log events

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# AWS X-Ray

AWS X-Ray

X-Ray traces requests that travel through your application and shows a map of your application's underlying component.

- **One-click** enablement for **Lambda functions** and integration with other AWS services
- **X-Ray SDK** to capture metadata for **API calls** that are made to AWS services with the **AWS SDK**
- Visual **detection** of **latency distribution** and quick **isolation** of **outliers** and trends
- Easier **debugging** of application code
- **Filtering** and **grouping** of requests by error type

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# Parts of an X-Ray trace

| Element | Description |
| --- | --- |
| Segments | Data about the work done by compute resources that run your application, including:<br>• Resource name<br>• Details about the request and the work that is done<br>• Subsegments |
| Subsegments | Breakdown of segment data into greater detail, including:<br>• More granular timing<br>• Details about downstream calls<br>• Inferred segments for downstream services<br>• Annotations and metadata |
| Annotations | Key-value pairs that can be indexed and used with filter expressions to group traces in the console for easier analysis |
| Metadata | Key-value pairs of any type that are not indexed but can be used to store data in the trace that you won't use for searching traces |

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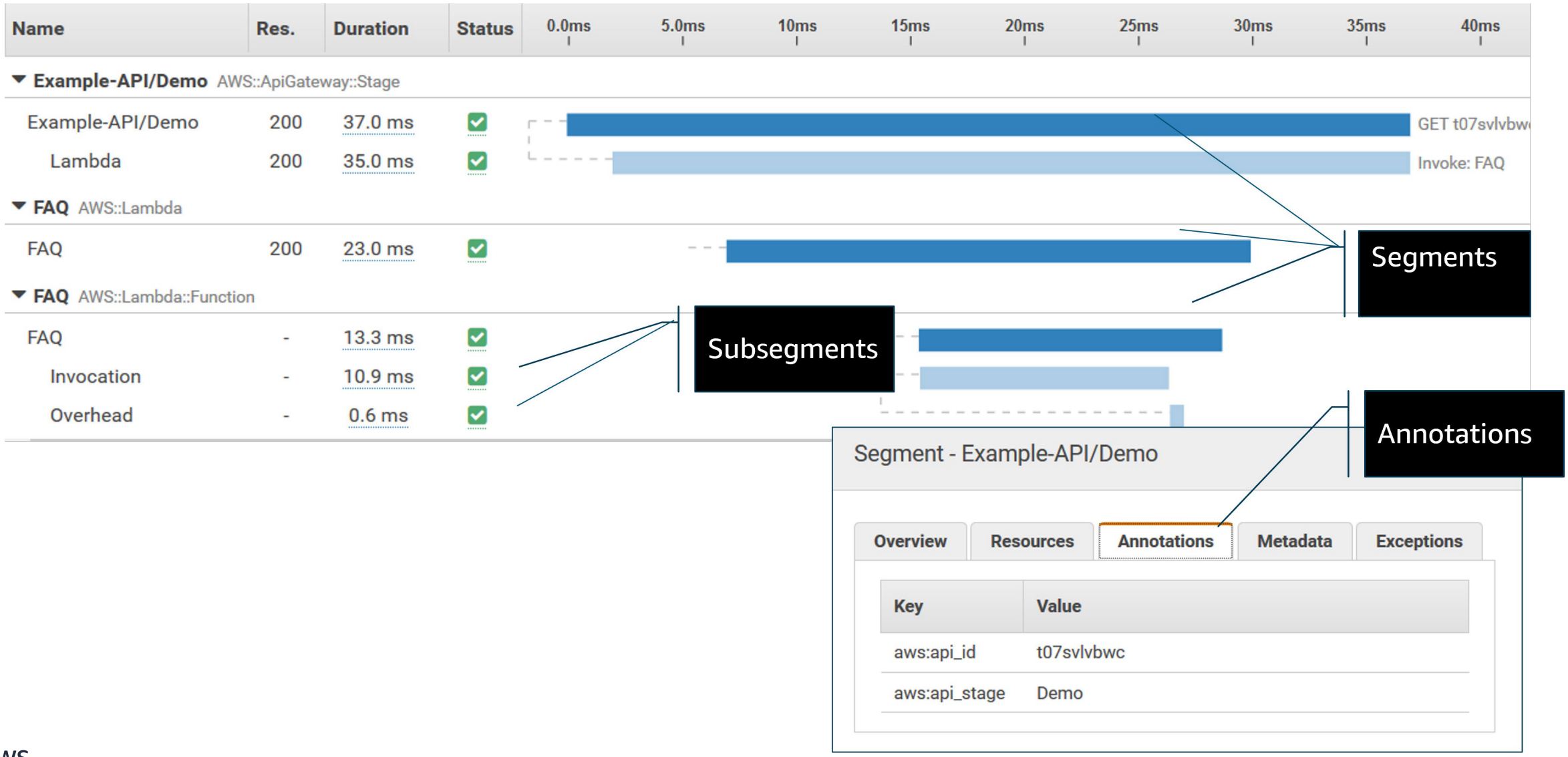
# Example X-Ray trace

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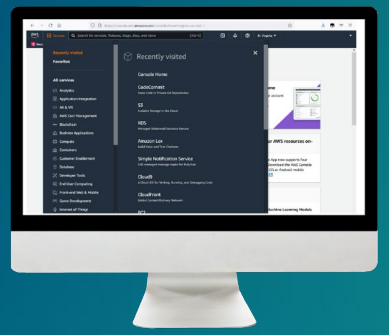
# Demonstration: Using X-Ray with Lambda



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# Section 8 key takeaways

- Modern applications require more than traditional operations monitoring.
- CloudWatch provides built-in logging and metrics to monitor your Lambda functions.
- With X-Ray, you can trace requests across your application to find trouble spots quickly.

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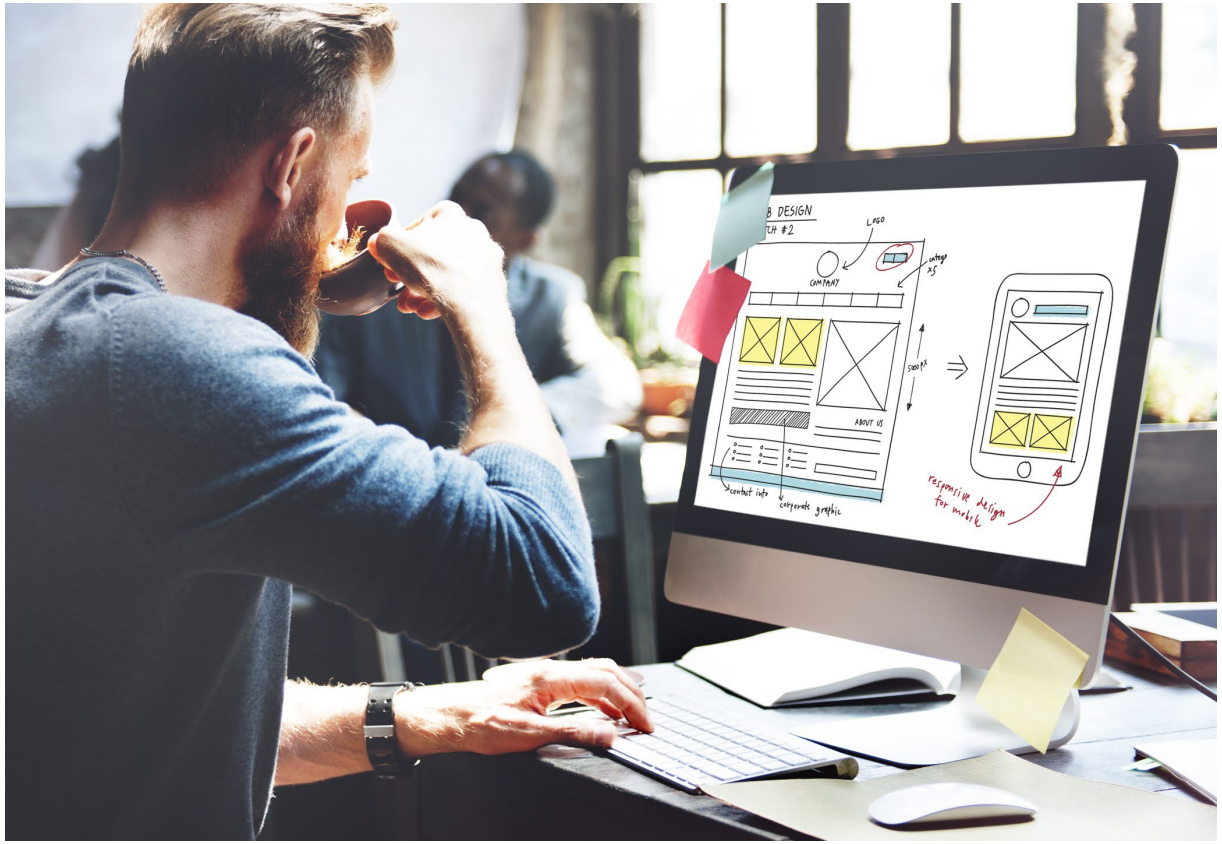
# Lab 7.1: Creating Lambda Functions Using the AWS SDK for Python



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# Lab: Scenario

In this lab, you again play the role of Sofía. You replace the mock endpoints that you created in the previous lab with real endpoints so that the web application can connect to the database. You will use Lambda to make this connection between the REST API and the data that is stored in DynamoDB.

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# Lab: Tasks

1. Preparing the development environment
2. Creating the get_all_products Lambda function
3. Configuring the REST API GET method to invoke the Lambda function
4. Creating the create_report Lambda function
5. Configuring the REST API POST method to invoke the Lambda function
6. Testing the integration by using the café website

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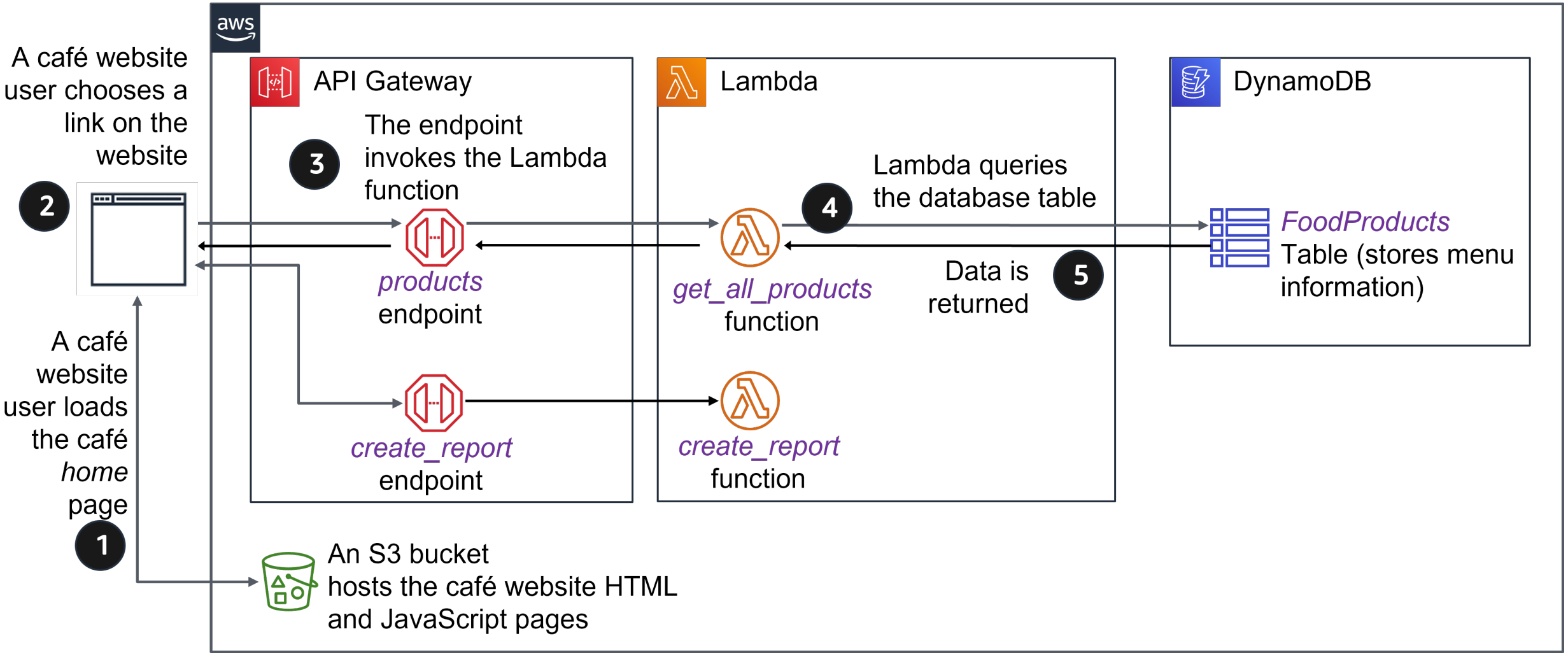
# Lab: Final product

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~ 90 minutes

# Begin Lab 7.1: Creating Lambda Functions Using the AWS SDK for Python

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# Lab debrief: Key takeaways

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# Module wrap-up

## Module 7: Developing Event-Driven Serverless Solutions

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# Module summary

In summary, in this module, you learned how to:

- Explain serverless computing
- Describe how Lambda works
- Recognize Lambda invocation models
- Identify how to use IAM to grant Lambda permissions
- Indicate the steps to author and configure a Lambda function
- Explain how to deploy Lambda functions
- Identify why X-Ray is a critical developer tool
- Create Lambda functions

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# Complete the knowledge check

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# Sample exam question

A developer is testing an application locally and has deployed it to Lambda. To remain under the package size limit, they did not include the dependencies in the deployment file. When testing the application remotely, the function does not run because of missing dependencies.

Which approach would best resolve the issue?

Identify the key words and phrases before continuing.

**The following are the key words and phrases:**

- **Under the package size limit**
- **Dependencies**

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# Sample exam question: Response choices

A developer is testing an application locally and has deployed it to Lambda. To remain **under the package size limit**, they did not include the **dependencies** in the deployment file. When testing the application remotely, the function does not run because of missing dependencies.

Which approach would best resolve the issue?

| Choice | Response |
|---|---|
| **A** | Use the Lambda console editor to update the code and include the missing dependencies. |
| **B** | Create an additional .zip file with the missing dependencies, and include the file in the original Lambda deployment package. |
| **C** | Add references to the missing dependencies in the Lambda function's environment variables. |
| **D** | Create a layer that contains the missing dependencies, and attach it to the Lambda function. |

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# Sample exam question: Answer

The correct answer is D.

| Choice | Response |
| --- | --- |
| **A** | Use the Lambda console editor to update the code and include the missing dependencies. |
| **B** | Create an additional .zip file with the missing dependencies, and include the file in the original Lambda deployment package. |
| **C** | Add references to the missing dependencies in the Lambda function’s environment variables. |
| **D** | **Create a layer that contains the missing dependencies, and attach it to the Lambda function.** |

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# Additional resources

- AWS Lambda Developer Guide:
  - Best Practices for Working with AWS Lambda Functions
  - Using AWS Lambda with Other Services
  - AWS Lambda Execution Environment
  - Troubleshooting Issues in Lambda
  - Monitoring Functions in the AWS Lambda Console
  - Accessing Amazon CloudWatch logs for AWS Lambda
  - Using AWS Lambda with AWS X-Ray
- Amazon CloudWatch Logs User Guide:
  - Analyzing Log Data with CloudWatch Logs Insights

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